

Anisotropic Superelliptic Shape Function (ASSF)

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Abstract

We present a single compact function that generates smooth n -dimensional shape masks with independent control over axis sizes, corner rounding, and anti-aliasing width. The formulation generalizes the superellipse by allowing per-axis exponent adjustment and adds a soft falloff for direct use in shaders. We provide a reference implementation.

Introduction

We created this function because commonly used shape functions target either purely rectangular shapes with circular corners, or emulate rounded shapes using vector graphics. Existing superellipse functions are common, but they often lack per-axis corner control and built-in anti-aliasing in a single expression. The ASSF fills that gap.

The function

$$f_{mask}(A, C, S, R, d, a) = 1 - \min\left(1, \sum_{n=1}^d \left| \frac{A_n - C_n}{S_n} \right|^{\frac{2S_n}{R_n}} \right)^{\frac{||R||}{a}}$$

Parameters:

- A: Sample point
- C: center point
- S: size radius
- R: corner pseudo-radius (superellipse approximation of a corner radius)
- d: working dimensions
- a: anti-aliasing radius

The function directly outputs a value from 0 to 1, representing the shape's opacity mask. A value of 1 means fully inside the shape, 0 fully outside, and intermediate values provide a smooth anti-aliased edge.

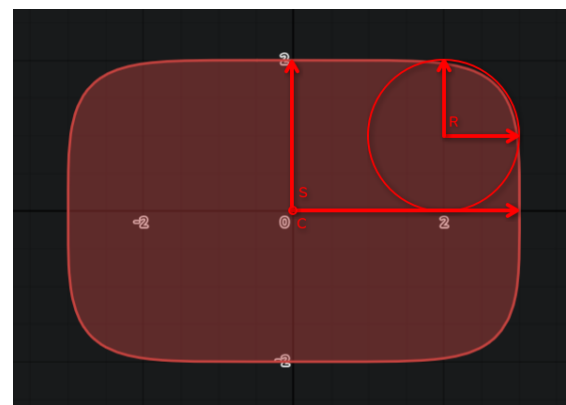


Figure 1: Demonstration of the shape function

Legend:

C: Center offset of (0,0)

S: Size radius of (2,3)

R: Corner radius of (1,1)

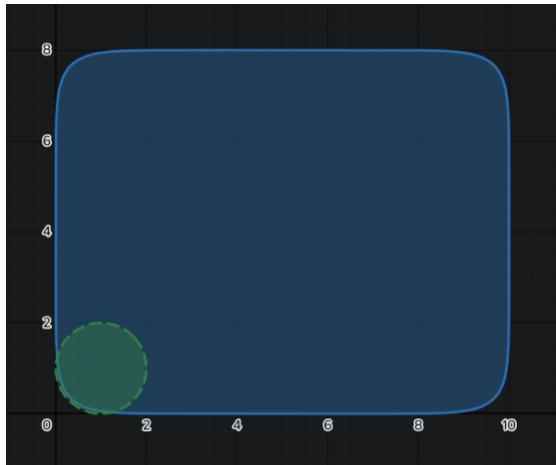


Figure 2: exponential corner smoothing

The underlying shape calculation function:

$$f_{\text{shape}}(A, C, S, R, d) = \sum_{n=1}^d \left| \frac{A_n - C_n}{S_n} \right|^{\frac{2S_n}{R_n}}$$

This is a modified p -norm: each axis is offset to move the origin to C , scaled by the size S , and raised to an axis-specific exponent $\frac{2S_n}{R_n}$. The exponent guarantees that the shape smoothly approaches a rounded corner of apparent radius R_n . The larger R_n , the lower the exponent and the rounder the corner. When all R_n are equal and all S_n are equal, the expression reduces to the classical superellipse.

Reference implementation

The following is a reference implementation of the ASSF in GLSL and its use.

```
float assf(vec2 sample, vec2
center, vec2 sizeRadius, vec2
```

```
cornerRadius, float
antialias) {
    vec2 u = abs((sample -
center) / sizeRadius);
    vec2 exponent = 2.0 *
sizeRadius / cornerRadius;
    float shape = pow(u.x,
exponent.x) + pow(u.y,
exponent.y);
    return 1.0 - pow(min(1.0,
shape), length(cornerRadius)
/ antialias);
}
```

Usage example:

```
float mask =
assf(gl_FragCoord.xy,
vec2(110.0, 90.0),
vec2(100.0, 80.0), vec2(20.0,
20.0), 2.0);
vec4 finalColor =
mix(background, foreground,
mask);
```

Sources and inspiration

Most of this math was manually rediscovered, but the following materials were used as inspiration and sources

[Gabriel Lamé - Wikipedia](#)

[Superellipse - Wikipedia](#)

[Pythagorean theorem - Wikipedia](#)

License

The ASSF and all accompanying code are released under the ISC License.

This license applies to the reference GLSL implementation and all code snippets contained within this document. The

*accompanying descriptive text and
mathematical notation are also released
under the same terms.*

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